

# Ecumenical Hyperdoctrines

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## Abstract

Prawitz’s Ecumenical system, as developed by Pereira, Pimentel and collaborators, places classical and intuitionistic logic side by side in a single natural deduction system, with each having its own implication while sharing conjunction and negation. We observe that the natural categorical semantics for this coexistence is given by what we call *Ecumenical hyperdoctrines*: fibred structures over a common base category in which each fibre carries both a Heyting algebra (modelling the intuitionistic fragment) and its Boolean reflection under double negation (modelling the classical fragment), connected by the  $\neg\neg$ -adjunction. This note makes the observation precise, establishes soundness and completeness, and discusses the relationship to Lawvere’s original hyperdoctrines and to the fibrational semantics of linear type theories. The relationship to Gödel doctrines and the proof-relevant generalisation of the construction are left as directions for future work.

## 1 Introduction

The question of how classical and intuitionistic logic can coexist in a single system without one being encoded into the other is the central theme of the Ecumenical programme initiated by Prawitz [1]. In Prawitz’s original natural deduction system, the two logics share conjunction and negation but each has its own implication:  $A \rightarrow B$  for the intuitionistic logician and  $A \Rightarrow B$  for the classical one. The key insight, made precise by Pereira and Rodriguez [2] and developed extensively by Pereira, Pimentel, and collaborators [3, 4, 5], is that the two implications have genuinely different meanings and neither should be defined in terms of the other. In particular, [4] shows that the Ecumenical perspective cannot be reduced to the Gödel–Gentzen translation, much less identified with it — a result that is directly relevant to our categorical construction, where the classical implication  $A \Rightarrow B = \neg\neg(A \rightarrow B)$  is modelled inside the Heyting algebra but is not definitionally equal to the intuitionistic implication.

From the proof-theoretic side, this programme is now well-developed: there are sequent calculi [3], cut elimination theorems [5], and modal extensions [6]. What is missing is a clean *categorical semantics* that respects the Ecumenical principle — that is, a semantics in which the classical and intuitionistic implications coexist as primitive operations, modelled by genuinely different categorical structures, neither encoding the other.

The obstacle to a naive categorical semantics is Joyal’s ‘obstruction’ observation: any cartesian closed category in which every object satisfies  $A \cong \neg\neg A$  must be a preorder. This rules out modelling classical propositional logic in a proof-relevant CCC in the most direct way. However, this obstruction applies to a *single* ambient CCC; it says nothing about *fibred* structures where classical content is localised in a subcategory rather than spread globally.

**A natural worry about negation.** A further obstacle, beyond Joyal’s theorem, initially suggested that even the posetal setting might not work. In Prawitz’s Ecumenical system, negation  $\neg A$  is *shared*: it has the same meaning for both the classical and intuitionistic logician, defined uniformly as  $A \rightarrow \perp$ . Since the classical implication satisfies  $A \Rightarrow B = \neg A \vee_c B$ , one might fear that sharing negation forces the two implications to collapse into one another — that there is simply no room for a genuinely separate classical implication once negation is fixed. The resolution is that in a Heyting algebra the  $\neg\neg$ -nucleus *preserves* negation: for any  $a \in H$ , the element  $\neg a$  is already  $\neg\neg$ -closed (since  $\neg\neg(\neg a) = \neg\neg\neg\neg a = \neg a$ ), so  $H^{\neg\neg}$  inherits negation from  $H$  without defining a new one. The shared negation is thus not an obstacle but a feature: it is precisely what guarantees that the Boolean reflection sits coherently inside the Heyting algebra, with the two implications distinguished while negation remains common to both. This observation, once made, is simple; but it is not obvious until one sees that  $\neg a \in H^{\neg\neg}$  for all  $a \in H$ .

The observation of this note is that hyperdoctrines — Lawvere’s fibred framework for predicate logic [7] — provide exactly the right setting. The key facts are:

- (i) A hyperdoctrine with **Heyting algebra fibres** models intuitionistic propositional logic.
- (ii) A hyperdoctrine with **Boolean algebra fibres** models classical propositional logic.
- (iii) The double negation map  $\neg\neg : H \rightarrow H$  on any Heyting algebra  $H$  defines a **nucleus** (a Lawvere–Tierney topology) whose fixed points form a Boolean algebra  $H^{\neg\neg}$ .
- (iv) This gives a natural **adjunction** between the Heyting and Boolean layers, fibrewise, without any global collapse.

An Ecumenical hyperdoctrine packages these four observations into a single fibred structure in which both implications are modelled directly and simultaneously. The intuitionistic implication  $A \rightarrow B$  is the Heyting implication in the full fibre; the classical implication  $A \Rightarrow B$  is the Boolean implication in the  $\neg\neg$ -closed subfibre. Neither encodes the other. We show in Section 5 that the Kripke semantics for the Ecumenical system [3, 4] is a special case of this construction: Ecumenical Kripke models are precisely the Ecumenical hyperdoctrines whose base is a poset and whose fibres are Heyting algebras of upward-closed sets.

**Relation to existing work.** Hyperdoctrines with Heyting fibres and with Boolean fibres are of course individually well known [7, 8]. What appears to be new is the combination — a single fibred structure carrying both layers simultaneously, with the  $\neg\neg$ -adjunction as the connecting map — motivated by and providing a categorical semantics for the Ecumenical system. We note the connection to the fibrational semantics of linear type theories in [11]: in that work, replacing cartesian fibres with symmetric monoidal closed fibres gives a semantics for linear logic; here, replacing Heyting fibres with their Boolean reflections gives a semantics for classical logic. The two constructions are parallel in structure and suggest a common generalisation.

**Plan of the note.** Section 2 recalls the propositional Ecumenical system. Section 3 recalls the relevant background on hyperdoctrines and the  $\neg\neg$ -nucleus. Section 4 gives the definition of Ecumenical hyperdoctrines and proves soundness. Section 5 presents examples. Section 6 discusses proof-relevant generalisations and future work.

## 2 The Propositional Ecumenical System

We recall the propositional fragment of Prawitz’s Ecumenical system, following Pereira and Pimentel [3].

**Definition 2.1** (Ecumenical propositional types). The types of the propositional Ecumenical system are generated by:

$$A ::= p \mid A \wedge B \mid \perp \mid \neg A \mid A \rightarrow B \mid A \Rightarrow B \mid A \vee_i B \mid A \vee_c B$$

where  $p$  ranges over propositional variables,  $\rightarrow$  is the *intuitionistic implication*,  $\Rightarrow$  is the *classical implication*,  $\vee_i$  is the *intuitionistic disjunction*, and  $\vee_c$  is the *classical disjunction*. Conjunction  $\wedge$ , negation  $\neg$ , and  $\perp$  are shared.

**Remark 2.2.** The classical implication can be defined as  $A \Rightarrow B := \neg A \vee_c B$ , or equivalently the classical logician’s implication satisfies  $A \Rightarrow B = \neg\neg(A \rightarrow B)$ . However, in the Ecumenical spirit we treat both as primitive. This is the key design choice: neither implication is defined in terms of the other.

The Ecumenical system has a natural deduction presentation in which:

- The **intuitionistic fragment** — the rules for  $\rightarrow, \wedge, \vee_i, \neg, \perp$  — is exactly the standard intuitionistic natural deduction system NJ.
- The **classical fragment** adds the rules for  $\Rightarrow$  and  $\vee_c$ , and in particular Peirce's law  $((A \Rightarrow B) \Rightarrow A) \Rightarrow A$  holds for  $\Rightarrow$  but not for  $\rightarrow$ .
- The **shared connectives**  $\wedge, \neg, \perp$  have the same rules in both fragments.

The key proof-theoretic property is that the Ecumenical system is **conservative** over both NJ and classical propositional logic: restricting to types built from  $\rightarrow$  alone gives exactly NJ, and restricting to types built from  $\Rightarrow$  alone gives exactly classical propositional logic.

### 3 Hyperdoctrines and Double Negation

We recall Lawvere's hyperdoctrines [7] in the propositional setting.

**Definition 3.1** (Propositional hyperdoctrine). A **propositional hyperdoctrine** is a functor  $P : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Pos}$  where  $\mathcal{C}$  is a category with finite products and each  $P(\Gamma)$  is a poset. We call  $P$  a **Heyting hyperdoctrine** if each  $P(\Gamma)$  is a Heyting algebra and each reindexing functor  $f^* : P(\Delta) \rightarrow P(\Gamma)$  preserves the Heyting structure. We call  $P$  a **Boolean hyperdoctrine** if each  $P(\Gamma)$  is a Boolean algebra.

Heyting hyperdoctrines model intuitionistic propositional logic and Boolean hyperdoctrines model classical propositional logic. This is entirely standard; see e.g. [8].

The key construction connecting the two is the double negation nucleus.

**Definition 3.2** (Double negation nucleus). Let  $H$  be a Heyting algebra with negation  $\neg$ . The **double negation map**  $\neg\neg : H \rightarrow H$  is a nucleus on  $H$ : it is inflationary ( $a \leq \neg\neg a$ ), idempotent ( $\neg\neg\neg\neg a = \neg\neg a$ ), and a meet-semilattice morphism ( $\neg\neg(a \wedge b) = \neg\neg a \wedge \neg\neg b$ ). The **Boolean reflection** of  $H$  is the set of  $\neg\neg$ -fixed points:

$$H^{\neg\neg} = \{a \in H \mid a = \neg\neg a\}$$

with the induced order.  $H^{\neg\neg}$  is a Boolean algebra, and the inclusion  $\iota : H^{\neg\neg} \hookrightarrow H$  has a left adjoint  $\neg\neg : H \rightarrow H^{\neg\neg}$ .

**Proposition 3.3.** *For any Heyting algebra  $H$ , the maps  $\neg\neg \dashv \iota : H^{\neg\neg} \hookrightarrow H$  form an adjunction, and  $H^{\neg\neg}$  is a Boolean algebra. Moreover, the Boolean implication in  $H^{\neg\neg}$  is given by  $a \Rightarrow b = \neg\neg(a \rightarrow b)$  for  $a, b \in H^{\neg\neg}$ .*

*Proof.* Standard; see e.g. [9]. The key point is that for  $a \in H^{\neg\neg}$  and  $b \in H^{\neg\neg}$ , the element  $\neg\neg(a \rightarrow b)$  is again  $\neg\neg$ -closed, and satisfies the universal property of the implication in  $H^{\neg\neg}$ .  $\square$

This proposition is the categorical heart of the observation: the classical implication  $a \Rightarrow b = \neg\neg(a \rightarrow b)$  lives naturally *inside* any Heyting algebra, in its Boolean reflection, without any additional structure. The two implications coexist in the same algebraic object, connected by the  $\neg\neg$ -adjunction.

## 4 Ecumenical Hyperdoctrines

We now define the categorical semantics for the Ecumenical system.

**Definition 4.1** (Ecumenical hyperdoctrine). An **Ecumenical hyperdoctrine** is a Heyting hyperdoctrine  $P : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Heyt}$  together with, for each object  $\Gamma \in \mathcal{C}$ , the Boolean reflection  $P(\Gamma)^{\neg\neg}$  of the fibre  $P(\Gamma)$ , such that:

- (i) **Fibrewise Boolean reflection.** For each  $\Gamma$ , the double negation map  $\neg\neg_{\Gamma} : P(\Gamma) \rightarrow P(\Gamma)^{\neg\neg}$  is natural in  $\Gamma$ : for every morphism  $f : \Delta \rightarrow \Gamma$  in  $\mathcal{C}$ , the reindexing functor  $f^*$  restricts to a map  $f^* : P(\Gamma)^{\neg\neg} \rightarrow P(\Delta)^{\neg\neg}$ , and the naturality square

$$\neg\neg_{\Delta} \circ f^* = f^* \circ \neg\neg_{\Gamma}$$

commutes.

- (ii) **Ecumenical separation.** The intuitionistic implication  $A \rightarrow B$  in  $P(\Gamma)$  and the classical implication  $A \Rightarrow B = \neg\neg(A \rightarrow B)$  in  $P(\Gamma)^{\neg\neg}$  are treated as distinct operations. In particular,  $A \Rightarrow B$  is not required to equal  $A \rightarrow B$  in  $P(\Gamma)$ .

We write  $\mathcal{E} = (P, P^{\neg\neg})$  for an Ecumenical hyperdoctrine.

**Remark 4.2.** The definition is deliberately minimal: an Ecumenical hyperdoctrine is nothing more than a Heyting hyperdoctrine together with the observation that its Boolean reflections vary naturally. No new structure is added; the classical layer is *already present* inside any Heyting hyperdoctrine, awaiting recognition. This is the main observation of this note.

We now give the interpretation of the Ecumenical propositional system in an Ecumenical hyperdoctrine.

**Definition 4.3** (Interpretation). Given an Ecumenical hyperdoctrine  $\mathcal{E} = (P, P^{\neg\neg})$  over  $\mathcal{C}$ , a context  $\Gamma \in \mathcal{C}$ , and an assignment of propositional variables

to elements of  $P(\Gamma)$ , we interpret types as follows:

$$\begin{array}{ll}
\llbracket p \rrbracket_\Gamma \in P(\Gamma) & \text{(given by the assignment)} \\
\llbracket A \wedge B \rrbracket_\Gamma = \llbracket A \rrbracket_\Gamma \wedge \llbracket B \rrbracket_\Gamma & \text{(in } P(\Gamma)) \\
\llbracket \neg A \rrbracket_\Gamma = \neg \llbracket A \rrbracket_\Gamma & \text{(in } P(\Gamma)) \\
\llbracket A \rightarrow B \rrbracket_\Gamma = \llbracket A \rrbracket_\Gamma \rightarrow \llbracket B \rrbracket_\Gamma & \text{(Heyting implication in } P(\Gamma)) \\
\llbracket A \Rightarrow B \rrbracket_\Gamma = \neg \neg (\llbracket A \rrbracket_\Gamma \rightarrow \llbracket B \rrbracket_\Gamma) & \text{(Boolean implication in } P(\Gamma)^{\neg\neg}) \\
\llbracket A \vee_i B \rrbracket_\Gamma = \llbracket A \rrbracket_\Gamma \vee \llbracket B \rrbracket_\Gamma & \text{(Heyting join in } P(\Gamma)) \\
\llbracket A \vee_c B \rrbracket_\Gamma = \neg \neg (\llbracket A \rrbracket_\Gamma \vee \llbracket B \rrbracket_\Gamma) & \text{(Boolean join in } P(\Gamma)^{\neg\neg})
\end{array}$$

A sequent  $\Gamma \vdash A$  is **valid** in  $\mathcal{E}$  if  $\llbracket A \rrbracket_\Gamma = \top$  in  $P(\Gamma)$ , and a classical sequent  $\Gamma \vdash_c A$  is valid if  $\llbracket A \rrbracket_\Gamma = \top$  in  $P(\Gamma)^{\neg\neg}$ .

**Theorem 4.4** (Soundness). *The interpretation of Definition 4.3 is sound for the propositional Ecumenical system: if  $\Gamma \vdash A$  is derivable in the Ecumenical system, then it is valid in every Ecumenical hyperdoctrine.*

*Proof.* We check each rule of the Ecumenical system. The rules for intuitionistic connectives are sound by the standard soundness theorem for Heyting hyperdoctrines. It remains to check the classical rules.

**Classical implication introduction.** If  $\Gamma, A \vdash_c B$  then  $\llbracket B \rrbracket = \top$  in  $P(\Gamma)^{\neg\neg}$  by induction. Since  $P(\Gamma)^{\neg\neg}$  is a Boolean algebra,  $\top \leq \llbracket A \rrbracket \Rightarrow \top = \top$ , so  $\llbracket A \Rightarrow B \rrbracket = \top$  in  $P(\Gamma)^{\neg\neg}$ .

**Peirce's law.** We must show  $((A \Rightarrow B) \Rightarrow A) \Rightarrow A$  is valid classically. In any Boolean algebra,  $((a \Rightarrow b) \Rightarrow a) \Rightarrow a = \top$  holds since Boolean algebras satisfy excluded middle: either  $a = \top$ , in which case the formula is trivially  $\top$ , or  $a = \perp$ , in which case  $a \Rightarrow b = \top$  and  $(a \Rightarrow b) \Rightarrow a = \perp \Rightarrow a = \top$ .

**Classical disjunction.**  $A \vee_c \neg A = \neg \neg (A \vee \neg A) = \neg \neg (\top) = \top$  in  $P(\Gamma)^{\neg\neg}$ , since  $A \vee \neg A = \top$  in the Heyting algebra and  $\neg \neg$  preserves  $\top$ .

All other cases are routine checks on the Heyting/Boolean algebra axioms.  $\square$

**Remark 4.5** (Ecumenical separation in the model). The soundness theorem validates the Ecumenical principle categorically: the classical rule  $(A \Rightarrow B) \Rightarrow A \vdash_c A$  (Peirce) is valid, but the intuitionistic version  $(A \rightarrow B) \rightarrow A \vdash A$  is not valid in a general Heyting algebra. The two implications behave differently in the model, as they should.

**Theorem 4.6** (Completeness). *The interpretation of Definition 4.3 is complete for the propositional Ecumenical system: if a sequent  $\Gamma \vdash A$  is valid in every Ecumenical hyperdoctrine, then it is derivable in the Ecumenical system. Similarly, if a classical sequent  $\Gamma \vdash_c A$  is valid in every Ecumenical hyperdoctrine, then it is derivable classically in the Ecumenical system.*

*Proof.* By the syntactic Ecumenical hyperdoctrine of Example 5.4. If  $\Gamma \vdash A$  is not derivable, then  $\llbracket A \rrbracket_\Gamma \neq \top$  in the syntactic model  $P_{\text{syn}}(\Gamma)$ , which is an Ecumenical hyperdoctrine by construction. Hence  $\Gamma \vdash A$  is not valid in every Ecumenical hyperdoctrine. The classical case is identical, using  $P_{\text{syn}}(\Gamma)^{\neg\neg}$ .  $\square$

**Remark 4.7** (Soundness and completeness, but not internal language). The soundness and completeness theorems together show that Ecumenical hyperdoctrines provide an adequate categorical semantics for the propositional Ecumenical system. However, they do *not* constitute an internal language theorem in the sense of [11]. An internal language theorem would require an equivalence of *categories* — between a category of Ecumenical theories and a category of Ecumenical hyperdoctrines — and this requires proof-relevant fibres in which morphisms, not just truth values, are tracked. In the posetal setting of this note, all parallel morphisms in each fibre are equal, so no such equivalence can be stated. Establishing an internal language theorem for the Ecumenical system is the central goal of the proof-relevant generalisation programme described in Section 6.

## 5 Examples

**Example 5.1** (The canonical example: open sets). Let  $X$  be a topological space and let  $\mathcal{C} = \{*\}$  be the one-object category. Define  $P(*) = \mathcal{O}(X)$ , the Heyting algebra of open sets of  $X$ , with Heyting implication  $U \rightarrow V = \text{int}(U^c \cup V)$ . The Boolean reflection is  $P(*)^{\neg\neg} = \{U \in \mathcal{O}(X) \mid U = \text{int}(\overline{U})\}$ , the regular open sets of  $X$ . The classical implication is  $U \Rightarrow V = \text{int}(\overline{U^c \cup V})$ . This is an Ecumenical hyperdoctrine. The Ecumenical separation is visible concretely:  $U \rightarrow V \neq U \Rightarrow V$  in general, since  $\text{int}(U^c \cup V)$  need not be regular open even when  $U$  and  $V$  are.

**Example 5.2** (Subobject classifier in a topos). Let  $\mathcal{T}$  be an elementary topos with subobject classifier  $\Omega$ . The subobject functor  $\text{Sub} : \mathcal{T}^{\text{op}} \rightarrow \mathbf{Heyt}$  is a Heyting hyperdoctrine [8]. The double negation  $\neg\neg : \Omega \rightarrow \Omega$  is a Lawvere–Tierney topology [9], and the  $\neg\neg$ -sheaves form a Boolean subtopos. The corresponding Ecumenical hyperdoctrine models the Ecumenical system inside any topos.

**Example 5.3** (Ecumenical Kripke models). The Kripke semantics for the Ecumenical system [3, 4] is a special case of our construction, and the  $\neg\neg$ -nucleus gives a clean account of the classical forcing condition.

Let  $(W, \leq)$  be a non-empty poset of **worlds** with a monotone **forcing relation**  $\Vdash$ . Take  $\mathcal{C}$  to be the poset category  $(W, \leq)$ . For each world  $w \in W$ , define  $P(w)$  to be the Heyting algebra of *upward-closed sets* of

worlds (propositions forced persistently from  $w$  onwards), with:

$$\begin{aligned}\llbracket A \wedge B \rrbracket_w &= \llbracket A \rrbracket_w \cap \llbracket B \rrbracket_w \\ \llbracket A \rightarrow B \rrbracket_w &= \{v \geq w \mid \forall u \geq v, u \in \llbracket A \rrbracket_w \Rightarrow u \in \llbracket B \rrbracket_w\}\end{aligned}$$

This is the standard intuitionistic Kripke forcing. The reindexing functor for  $w \leq w'$  is restriction to upsets above  $w'$ , which preserves the Heyting structure.

The Boolean reflection  $P(w)^{\neg\neg}$  consists of the **regular** upward-closed sets: those  $U$  such that  $U = \neg\neg U$ , i.e. those not falsifiable at any future world. The classical implication is then:

$$\llbracket A \Rightarrow B \rrbracket_w = \neg\neg(\llbracket A \rightarrow B \rrbracket_w) = \{v \geq w \mid \neg\neg(v \in \llbracket A \rightarrow B \rrbracket_w)\}$$

This is precisely the classical forcing condition in the Ecumenical Kripke semantics:  $A \Rightarrow B$  is forced at  $w$  if it is not possible that  $A$  is forced while  $B$  is not, in the double negation sense. The Ecumenical separation is immediate:  $A \rightarrow B$  is an upset (forced persistently) while  $A \Rightarrow B$  is its regular closure, and these coincide only when the model is already Boolean (i.e. when excluded middle holds for  $A \rightarrow B$ ).

This example shows that Ecumenical Kripke models are precisely the Ecumenical hyperdoctrines whose base is a poset and whose fibres are Heyting algebras of upward-closed sets. The categorical framework thus subsumes and generalises the Kripke semantics.

**Example 5.4** (Syntactic example). The Lindenbaum–Tarski algebra of the propositional Ecumenical system yields the canonical syntactic Ecumenical hyperdoctrine, in which:

- $P(\Gamma)$  is the Heyting algebra of equivalence classes of intuitionistic propositions under  $\Gamma \vdash A = B$ .
- $P(\Gamma)^{\neg\neg}$  is the Boolean algebra of equivalence classes of classical propositions under  $\Gamma \vdash_c A = B$ .
- The  $\neg\neg$ -map sends  $[A]$  to  $[\neg\neg A]$ .

Completeness for the Ecumenical system reduces to showing this is an Ecumenical hyperdoctrine, which follows immediately from the definitions.

## 6 Discussion and Future Work

**Why this is the right semantics.** The definition of an Ecumenical hyperdoctrine is deliberately parsimonious: it adds no new structure to a Heyting hyperdoctrine, but simply recognises the Boolean reflection that is already present. This matches the Ecumenical proof-theoretic philosophy:



the classical fragment is not an extension of the intuitionistic one, nor does the intuitionistic fragment encode the classical one; they coexist as two aspects of a single structure. The  $\neg\neg$ -adjunction  $\neg\neg \dashv \iota : P(\Gamma)^{\neg\neg} \hookrightarrow P(\Gamma)$  plays the same structural role as the modality adjunction in linear logic: it connects two layers without collapsing one into the other. The fact that Ecumenical Kripke models (Example 5.3) are a special case provides independent confirmation: the classical forcing condition in the Kripke semantics is exactly the  $\neg\neg$ -closure of the intuitionistic forcing condition, which is precisely what our construction computes fibrewise.

**Joyal’s obstruction is avoided.** Joyal’s theorem applies to a CCC in which every object is  $\neg\neg$ -closed. In our setting,  $\neg\neg$ -closed objects form a *substructure* of the fibre, not the whole fibre. The proof-relevant structure, when we lift to it, will live in the Heyting layer; the Boolean layer remains posetal. This is the key move that avoids the obstruction.

**The proof-relevant generalisation.** The main limitation of this note is that hyperdoctrines have *posetal* fibres, making them proof-irrelevant. For a type theory — in particular for the Ecumenical type theory implicit in the Curry–Howard reading of the Ecumenical system — one needs proof-relevant fibres. The natural programme is:

- (i) Replace Heyting algebra fibres with **cartesian closed categories** (modelling the intuitionistic fragment proof-relevantly), following the standard lifting from Heyting hyperdoctrines to locally cartesian closed categories.
- (ii) Identify the  $\neg\neg$ -closed objects in each fibre as a **reflective subcategory**, and ask when this subcategory is again cartesian closed.
- (iii) Find proof-relevant models in which the  $\neg\neg$ -reflection is well-behaved. We note explicitly that the **Dialectica construction**  $\text{Dial}(\mathcal{C})$  does *not* directly apply here:  $\text{Dial}(\mathcal{C})$  models **classical linear logic** (giving rise to star-autonomous categories), not classical propositional logic in the intuitionistic/classical sense required by the Ecumenical system. The  $\neg\neg$  of classical linear logic — where  $\neg\neg A = A$  holds by involutive negation — is a fundamentally different operation from the  $\neg\neg$ -nucleus of a Heyting algebra, which identifies only a proper substructure of Boolean-closed elements. The search for appropriate proof-relevant models remains open.

We leave this programme for future work, noting that the fibrational methods developed for linear type theories in [11] provide the right technical framework, even if the specific constructions need to be developed anew.

**Adding linearity: towards a full Ecumenical system with three implications.** Once both the intuitionistic and classical layers are modelled proof-relevantly, a linear implication  $A \multimap B$  can be added as a third primitive by replacing the CCC fibres with symmetric monoidal closed categories, following the fibrational approach to the Intuitionistic and Linear Type Theory (ILT) of [11]. In ILT, the base category models intuitionistic contexts and the fibres — which are symmetric monoidal closed rather than cartesian — model linear terms parametrised by those contexts. Combining this with the present construction would yield a fibred structure with three layers:

classical (Boolean reflection)  $\subset$  intuitionistic (CCC fibres)  $\supset$  linear (SMC fibres)

connected by the  $\neg\neg$ -adjunction (classical/intuitionistic) and the comprehension adjunction (intuitionistic/linear). The resulting system — a type theory with three coexisting implications, none encoding either of the others — is the subject of ongoing work by the author, of which this note is the first step.

**Relation to the Gödel doctrines programme.** The Gödel doctrines of [10] characterise the Dialectica construction fibrationally via the logical principles of Skolemization and prenex normal form: a fibration is an instance of the Dialectica construction if and only if it is a Gödel fibration. Ecumenical hyperdoctrines are in some sense a companion construction: rather than capturing the Dialectica construction through Skolemization, they capture the coexistence of classical and intuitionistic logic through the  $\neg\neg$ -nucleus. Both use the fibrational language to identify a classical layer inside an intuitionistic structure, but the mechanisms and the logical principles they embody are quite different. The precise relationship between the two constructions is an interesting open question.

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